LLM Course Final Project

AI Newsroom — how far does multi-agent decomposition + self-correction take a small LM?

A small open-source LM, split into role agents — can it close the gap to a large LLM?
A measured study, not a product.

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Background

SLMs are cheap but weak → can structure + self-correction close the gap?

SLM (small LM) cheap · runs locally → but weaker than large LLMs

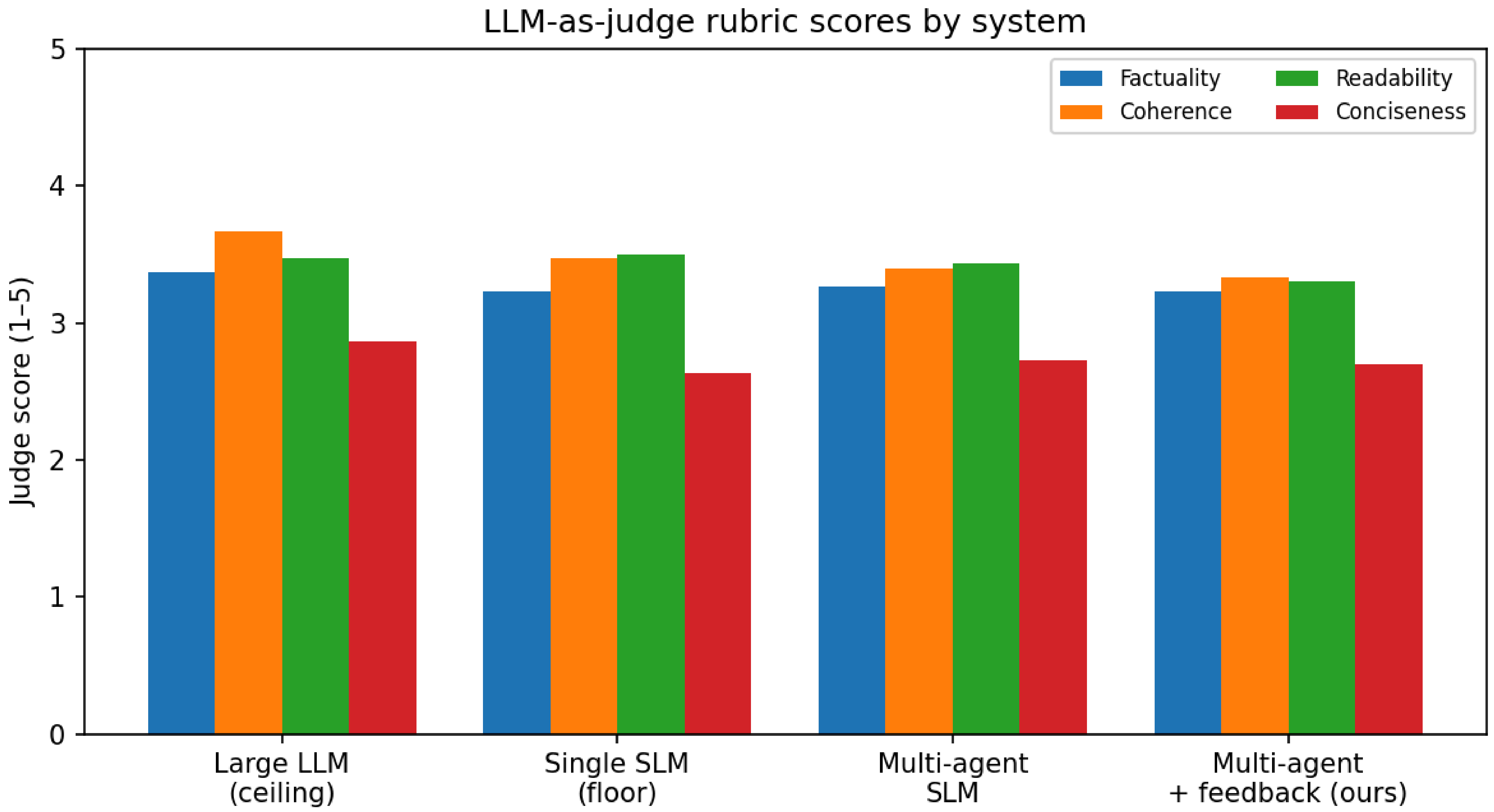
Question Does role decomposition + self-correction recover large-LLM quality — and at what cost?

Testbed automated newsletter generation (fuse 5 source articles → 1 article)

Core idea: instead of just scaling the model, measure whether organizing a small model lifts quality.

System

Four role agents form a newsroom → an Editor that sends it back: feedback loop



Scout collect
→ **Reader** parallel 3-line summaries
→ **Writer** fuse into one article
→ **Editor** score + feedback

Editor → Writer feedback loop (K ≤ 2)
re-write if below threshold
- the "new method" under study

Setup

Four systems compared on identical frozen inputs

Systems
api Gemini-3.5-flash (ceiling)
single_slm Llama-3.2-3B (floor)
multi_agent decomposition only
multi_agent_fb + feedback loop

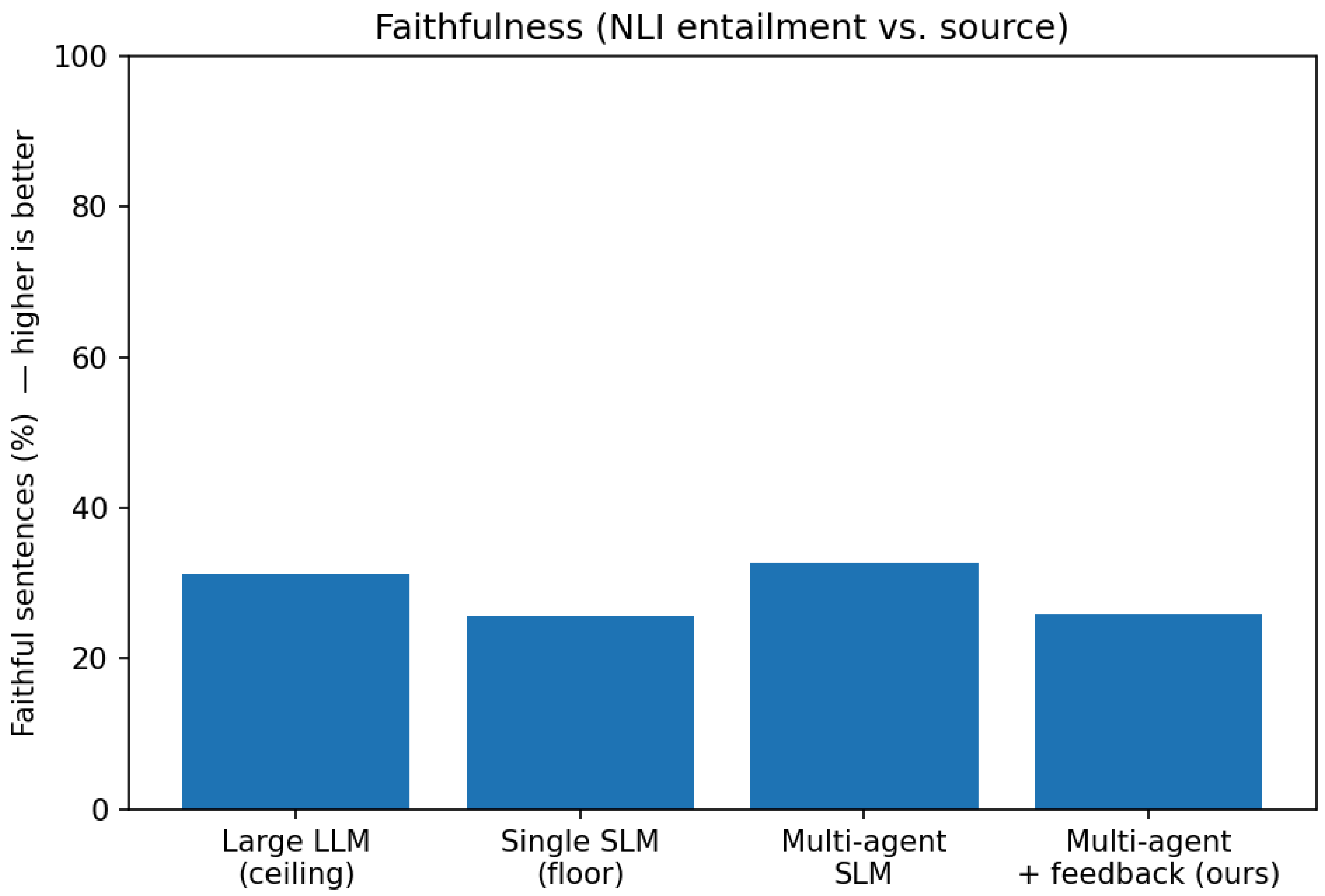
Data
HackerNews (tech) + arXiv (research)
30 newsletters (n = 30)

Metrics
LLM-judge rubric (4 axes)
pairwise win-rate
NLI faithfulness (judge-free)
latency / iters

Bias control
judge = independent **Qwen2.5-14B**
≠ generator / ceiling family
→ less self-preference
pairwise order swapped
→ removes position bias

Key Results

Decomposition raises faithfulness — past the API ceiling ✅



single_slm **25.6%**
→ multi_agent **32.7%** faithful

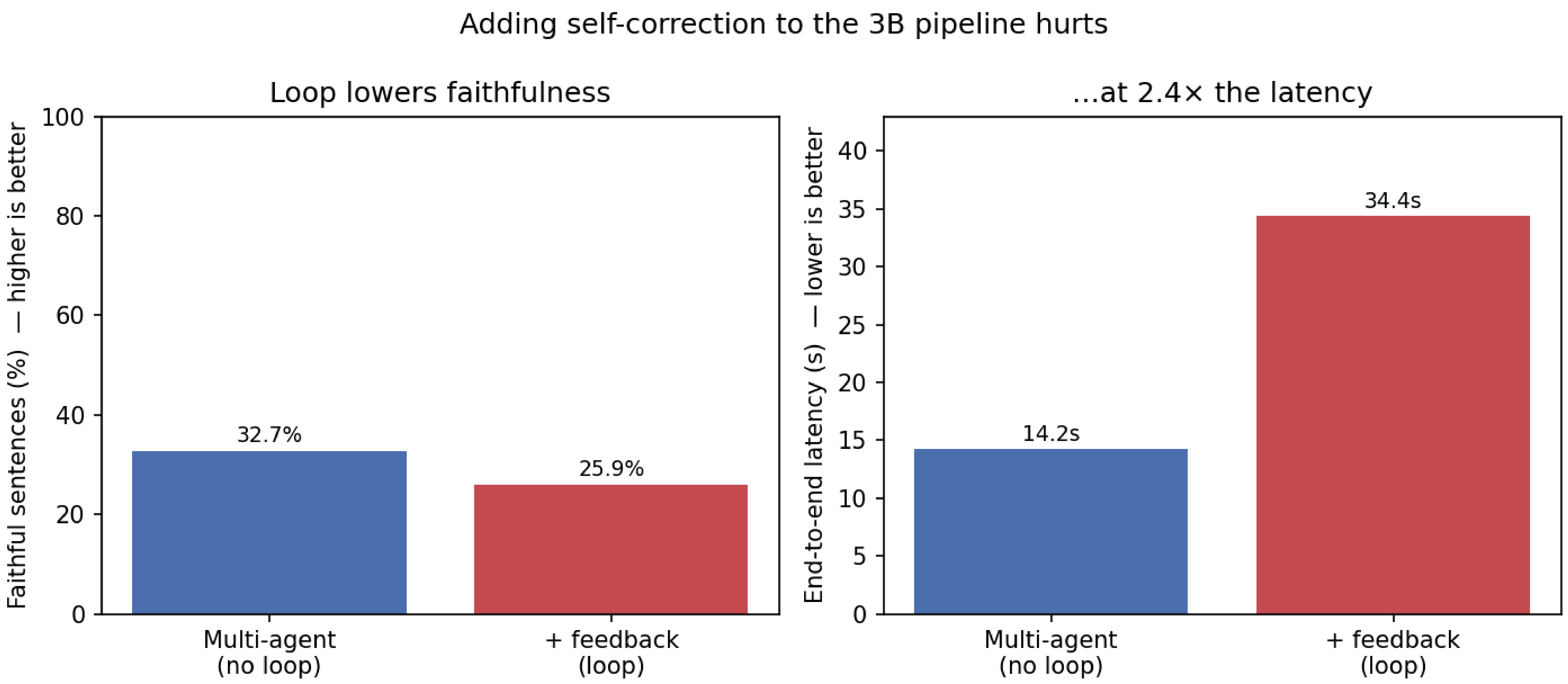
multi_agent 32.7% > API ceiling 31.2%

Reader's per-article summaries
cut hallucination vs one-shot fusion

Splitting the work so each part stays faithful, then merging, can beat a large model that fuses in one shot.

Key Results

Self-correction backfires 😢



With the loop
faithful 32.7% → **25.9%** (back to floor)

judge 3.21 → 3.14
latency **2.4×** (14s → 34s)

The 3B reviser can't act on its own critique
→ revisions drift / degrade

Key Results

All systems × metrics — full scoreboard (scores_table)

Table 1: Newsletter quality, faithfulness, and cost across four systems ($n = 30$).

System	Judge avg.	Faithful (%)	Win rate (%)	Latency (s)	Iters
API (ceiling)	**3.34**	31.2	**66.7**	14.02	0
Single SLM (floor)	3.21	25.6	–	**9.17**	0
Multi-agent	3.21	**32.7**	13.3	14.24	0
+ feedback (ours)	3.14	25.9	20.0	34.40	1.87

API is a clear ceiling (judge 3.34 · win-rate 66.7%)
judge_avg spread is small → read faithfulness + win-rate first

legend — judge_avg: mean of 4-axis rubric · faithfulness: NLI entailment rate · win-rate: pairwise wins

Discussion

Self-correction's payoff is gated by base-model capability

Key insight
decomposition (structural) → helps
iterative self-revision → capability-dependent
no help at 3B

A weak model is a weak self-critic.

Limitations

- n = 30 · single judge
- Small judge_avg spread (no significance test)
- Faithfulness is an NLI proxy
- Conciseness is everyone's weakest axis (task-inherent)

Future

- stronger / separate reviser
- per-agent specialized models
- human eval

Reproducibility & Wrap-up

Anyone can reproduce end-to-end on identical inputs

Free official APIs (HackerNews · arXiv) · eval set frozen → all systems on identical inputs

One config.yaml swaps any model

python main.py all reproduces end-to-end

Takeaway

Decomposition lifts a small LM's faithfulness above a large-LLM ceiling**.**
But naive self-correction needs a capable base model**.**

Final Thoughts

What works today vs what becomes possible

Now
Structural decomposition pays off even with small models.

Drops straight into local, low-cost pipelines — "role division" alone lifts quality.

Future
For self-correction to work, the reviser must cross a capability threshold.

As models grow, the feedback loop becomes an asset, not a cost.

Thank you.